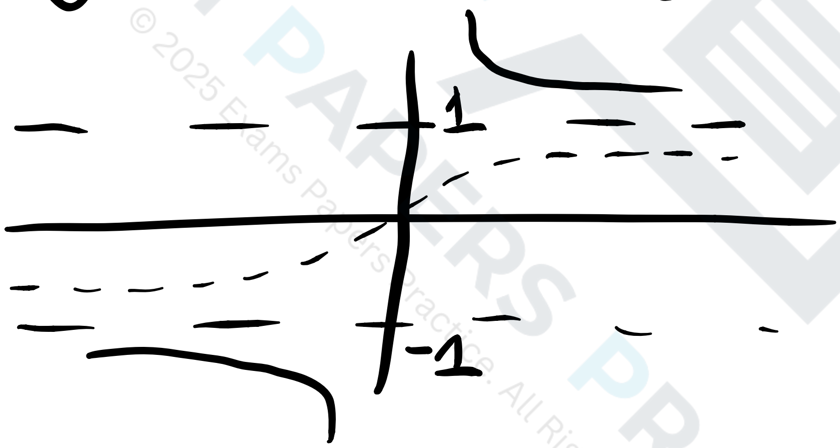
Differentiating hyperbolic trig functions

Basic skills

Q1. $\sinh 2x = \dfrac{e^{2x} - e^{-2x}}{2}$ $\qquad \sinh x = \dfrac{e^{x} - e^{-x}}{2}$

Q2. $y = \coth x = \dfrac{1}{\tanh x}$



Q3. $\dfrac{d}{dx}\left(e^{kx}\right) = ke^{kx}$

Q4. $y = 5\sinh x \Rightarrow \dfrac{dy}{dx} = 5\cosh x$

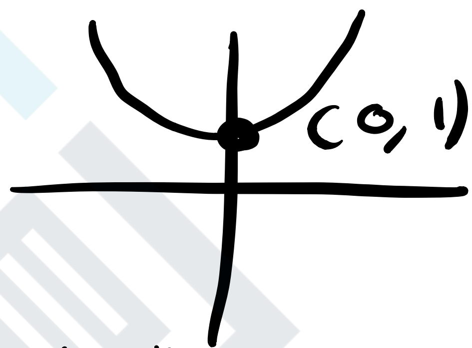
so $\dfrac{dy}{dx}(0) = 5\cosh 0 = 5$

so grad is 5

Q5. $y = \tanh 3x \Rightarrow \dfrac{dy}{dx} = 3\,\text{sech}^2 3x$

so $\dfrac{dy}{dx}(2) = 3\,\text{sech}^2 6 = \dfrac{12}{(e^6+e^{-6})^2}$

Q6. $y = \cosh x$



$\dfrac{dy}{dx} = \sinh x$

$\sinh x = 0$ at $x = 0$ stationary

could use inverse
$x = \sinh^{-1} 0$
$= \ln(0+\sqrt{0+1}) = \ln 1$
$= 0$

$\dfrac{d^2y}{dx^2} = \cosh x$

$\cosh 0 = 1 > 0$ so min

Q7. $\dfrac{d}{dx}(\cosh x) = \dfrac{d}{dx}\left(\dfrac{e^x + e^{-x}}{2}\right)$

$= \dfrac{e^x - e^{-x}}{2} = \sinh x$

Competency skills

Q 8. a) $\frac{d}{dx}(\tanh x) = \frac{d}{dx}\left(\frac{\sinh x}{\cosh x}\right)$

Quotient rule

$u = \sinh x$ $v = \cosh x$

$u' = \cosh x$ $v' = \sinh x$

$$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x}$$

$\cosh^2 x - \sinh^2 x = 1$

$$= \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x$$

b) $\frac{d}{dx}(\tanh x) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)$

Quotient rule

$u = e^x - e^{-x}$ $v = e^x + e^{-x}$

$u' = e^x + e^{-x}$ $v' = e^x - e^{-x}$

$$= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= \frac{e^{2x} + 2 + e^{-2x} - (e^{2x} - 2 + e^{-2x})}{(e^x + e^{-x})^2}$$

$$= \frac{4}{(e^x + e^{-x})^2} = \left(\frac{2}{e^x + e^{-x}}\right)^2 = \operatorname{sech}^2 x$$

Q9.

a) $x = \sinh y$ want $\frac{dy}{dx}$

$\frac{dx}{dy} = \cosh y$ now $\cosh^2 y = 1 + \sinh^2 y$

so $\frac{dx}{dy} = \sqrt{1+\sinh^2 y} = \sqrt{1+x^2}$

so $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\sqrt{1+x^2}}$

b) $\frac{d}{dx}(\sinh^{-1} x) = \frac{d}{dx}\left(\ln(\underbrace{x+\sqrt{x^2+1}}_{u})\right)$

chain rule (sub)

$= \frac{d}{du}(\ln u) \times \frac{d}{dx}\left(x+\sqrt{x^2+1}\right)$

chain $\sqrt{u}$ $u = x^2+1$

$\frac{1}{2\sqrt{u}} \times 2x$

$= \frac{1}{u} \times \left(1 + x(x^2+1)^{-\frac{1}{2}}\right)$

$= \frac{1 + \frac{x}{\sqrt{x^2+1}}}{x+\sqrt{x^2+1}} = \frac{\sqrt{x^2+1}+x}{x\sqrt{x^2+1}+x^2+1}$

$= \frac{x+\sqrt{x^2+1}}{(x+\sqrt{x^2+1})\sqrt{x^2+1}} = \frac{1}{\sqrt{x^2+1}}$

Q10. $y = \sinh x \cosh x = \frac{1}{2} \sinh 2x$

$\sinh 2x = 2 \sinh x \cosh x$

So $\frac{dy}{dx} = 2 \times \frac{1}{2} \cosh 2x = \cosh 2x$

$\cosh 2x \neq 0$ as $\cosh 2x \geq 1$.

So no stationary points

Q11. $f(x) = \operatorname{sech} x = \frac{1}{\cosh x}$

Quotient rule
$u = 1 \quad v = \cosh x$
$u' = 0 \quad v' = \sinh x$

so $f'(x) = \frac{-\sinh x}{\cosh^2 x} = -\tanh x \operatorname{sech} x$

so $f'(1) = -\tanh 1 \operatorname{sech} 1$

$f(1) = \operatorname{sech} 1$

so tangent $y = -x \tanh 1 \operatorname{sech} 1 + c$

$\operatorname{sech} 1 = -\tanh 1 \operatorname{sech} 1 + c$

$\Rightarrow c = (\tanh 1 + 1) \operatorname{sech} 1$

so $y = -x \tanh 1 \operatorname{sech} 1 + (\tanh 1 + 1) \operatorname{sech} 1$

is the tangent

Q 11. For $\frac{d}{dx}\left(\operatorname{arccosh}\left(\frac{x}{a}\right)\right)$ let

$y = \operatorname{arccosh}\left(\frac{x}{a}\right)$ and want $\frac{dy}{dx}$

so $x = a\cosh y$

$\cosh^2 x - \sinh^2 x = 1$

$$\Rightarrow \frac{dx}{dy} = a\sinh y = a\sqrt{\cosh^2 y - 1}$$

$$= \sqrt{a^2\cosh^2 y - a^2} = \sqrt{x^2 - a^2}$$

so $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - a^2}}$

Q 12. For $\frac{d}{dx}\left(\arctan\left(\frac{x}{a}\right)\right)$ find $\frac{dy}{dx}$ with

$y = \operatorname{arctanh}\left(\frac{x}{a}\right) \Rightarrow x = a\tanh y$

$\operatorname{sech}^2 x + \tanh^2 x = 1$

$$\text{so } \frac{dx}{dy} = a\operatorname{sech}^2 y = a(1 - \tanh^2 y)$$

$$= a - a\tanh^2 y = a - a\left(\frac{x}{a}\right)^2$$

$$= a - \frac{x^2}{a} = \frac{a^2 - x^2}{a}$$

so $\frac{dy}{dx} = \frac{a}{a^2 - x^2}$

Advanced skills

Q14. $f(x) = \operatorname{arcsinh}(x^2)$

$f(u) = \operatorname{arcsinh}(u) \qquad u = x^2$

$$f'(x) = \frac{df}{du}\frac{du}{dx} = \frac{1}{\sqrt{u^2+1}} \times 2x = \frac{2x}{\sqrt{x^4+1}}$$

as $\frac{d}{dx}\left(\operatorname{arcsinh}(x)\right) = \frac{1}{\sqrt{x^2+1}}$

Q15. $f(x) = e^{kx}\cosh x$

$f'(x) = ke^{kx}\cosh x + e^{kx}\sinh x$

$= e^{kx}(k\cosh x + \sinh x)$

for $f'(x) = 0 \Rightarrow k\cosh x + \sinh x = 0$

$\Rightarrow k + \tanh x = 0 \qquad \tanh x = -k$

as $-1 < \tanh x < 1 \qquad \Rightarrow -1 < -k < 1$

so $-1 < k < 1$

Q16. $y = x^2 \sinh x^2 \qquad y = u\sinh u,\ u = x^2$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = (u\cosh u + \sinh u) \times 2x$

so $\dfrac{dy}{dx} = 2x\left(x^2\cosh x^2 + \sinh x^2\right)$

$= 2x^3\cosh x^2 + 2x\sinh x^2$

Q17. $y = \text{arc sech}\left(\dfrac{x}{4}\right)$

a) $x = 4\,\text{sech}\,y$ now $\dfrac{dx}{dy} = -4\,\text{sech}\,y\tanh y$

$\dfrac{d}{dx}(\text{sech}\,x) = -\text{sech}\,x\tanh x$
use quotient rule to see

so $\dfrac{dx}{dy} = -x\tanh y$

$\text{sech}^2 x + \tanh^2 x = 1$

$= -x\sqrt{1-\text{sech}^2 y}$

$= -x\sqrt{1-\left(\dfrac{x}{4}\right)^2} = -\dfrac{1}{4}x\sqrt{16-x^2}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{-4}{x\sqrt{16-x^2}}$

b) $\displaystyle\int_1^2 \frac{5}{x\sqrt{64-4x^2}}\,dx = \int_1^2 -\frac{5}{16}\times\frac{-4}{x\sqrt{16-x^2}}\,dx$

$= -\dfrac{5}{16}\left[\text{arc sech}\left(\dfrac{x}{4}\right)\right]_1^2 = -\dfrac{5}{16}\left(\text{arc sech}\left(\tfrac{1}{2}\right) - \text{arcsech}(1)\right)$

as reciprocal functions

$= -\dfrac{5}{16}\left(\text{arc cosh}(2) - \text{arc cosh}(1)\right)$

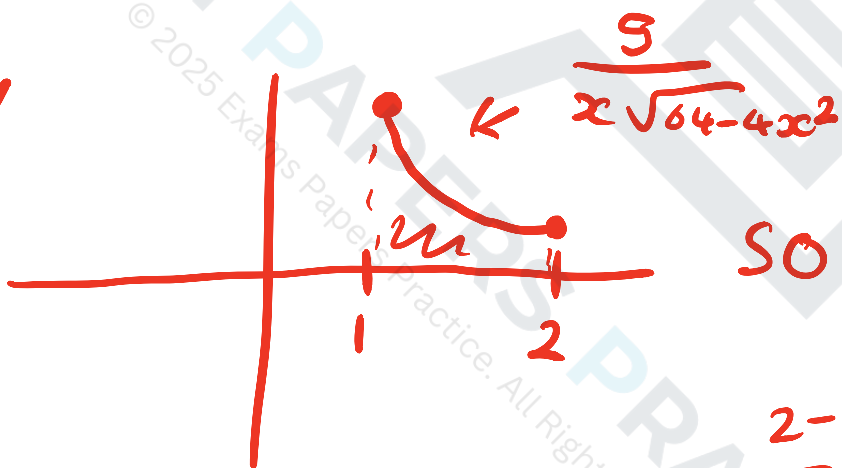
$$= -\frac{5}{16}\left(\ln\left(2 \pm \sqrt{2^2-1}\right) - \ln\left(1 \pm \sqrt{1^2-1}\right)\right)$$

$$= -\frac{5}{16}\ln\left(2 \pm \sqrt{3}\right)$$

Question: why two answers?

as $\operatorname{arccosh}(2)$ has two values.

Now



So Positive Area

$2-\sqrt{3} < 1$

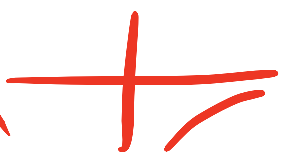
now only $-\frac{5}{16}\ln\left(2-\sqrt{3}\right)$ is Positive

so this is the answer

Negative $-\frac{5}{16}\ln\left(2+\sqrt{3}\right) = \frac{5}{16}\ln\left(\frac{1}{2+\sqrt{3}}\right)$

$= \frac{5}{16}\ln\left(2-\sqrt{3}\right) < 0$

is from neg version



i.e $\int_1^2 \frac{-5}{x\sqrt{64-4x^2}}\,dx$